

Human review packet: LBG24SpatialUnderreporting

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: LBG24SpatialUnderreporting
Paper id: LBG24SpatialUnderreporting
Source version: not recorded
Source record: audit/paper_statement_map.json
Rows: 4
Generated: 2026-08-18
Source URL: [official source](#)

Review status: 0/4 source-claim screens; 0/38 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper LBG24SpatialUnderreporting --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

Contents

- Semantic library prerequisites (38; not paper claims)
 - [EconCSLib.Probability.Exponential.Model](#)
 - [EconCSLib.Probability.Exponential.Model.rate](#)
 - [EconCSLib.Probability.Exponential.Model.measure](#)
 - [EconCSLib.Probability.Exponential.Model.mk](#)
 - [EconCSLib.Probability.PoissonCountingMeasureByLaw](#)
 - [EconCSLib.Probability.PoissonCountingMeasureByLaw.finiteCount](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.count](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.intervalCount](#)
 - [EconCSLib.Probability.PoissonProcess.rateExposureParam](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.mk](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.rate](#)
 - [EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.unitIntervalCount](#)
 - [EconCSLib.Probability.PoissonProcess.ObservationWindow](#)
 - [EconCSLib.Probability.PoissonProcess.ObservationWindow.endTime](#)
 - [EconCSLib.Probability.PoissonProcess.ObservationWindow.startTime](#)
 - [EconCSLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.count](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.jumpTime](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.window](#)
 - [EconCSLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)
 - [EconCSLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
 - [EconCSLib.Probability.PoissonProcess.terminalTailFromJumpTimes](#)
 - [EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail](#)
 - [EconCSLib.Probability.PoissonProcess.interarrival](#)

- EconCSLib.Probability.PoissonProcess.arrivalTime
- EconCSLib.Probability.PoissonProcess.canonicalRenewalCount
- EconCSLib.Probability.PoissonProcess.exponentialInterarrivalMeasure
- EconCSLib.Probability.PoissonProcess.canonicalForwardCount_hasIndepIncrements
- EconCSLib.Probability.PoissonProcess.isProbabilityMeasure_exponentialInterarrivalMeasure
- EconCSLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw
- EconCSLib.Probability.PoissonProcess.countLikelihood
- EconCSLib.Probability.PoissonProcess.cumulativeArrivalVector
- EconCSLib.Probability.PoissonProcess.exponentialBlockDensity
- EconCSLib.Probability.PoissonProcess.forwardPostStopIntervalCount
- EconCSLib.Probability.PoissonProcess.noArrivalProb
- Source claims (4)
 - appendixTheorem2Eq8CausalStoppingSourceSpec
 - lemma1CalendarTimeDurationSourceObservedProcessSpec
 - proposition1CalendarTimeHomogeneousSourceSpec
 - lemma2ForwardSelectedStartExponentialTailSpec

Material library prerequisites

EconCSLib.Probability.Exponential.Model

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type

Exact Lean library declaration

```
structure Model where
  rate : ℝ
  rate_pos : 0 < rate
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.Exponential.Model.rate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

`(self : EconCSLib.Probability.Exponential.Model) → self.1`

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.Exponential.Model.measure

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(M : EconCSLib.Probability.Exponential.Model) → ProbabilityTheory.expMeasure M.rate

Exact Lean library declaration

```
noncomputable def measure (M : Model) : MeasureTheory.Measure ℝ :=  
  ProbabilityTheory.expMeasure M.rate
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.Exponential.Model.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

(rate : $\mathbb{R}$) $\rightarrow$ $0 < \text{rate} \rightarrow \text{EconCSLib.Probability.Exponential.Model}$

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonCountingMeasureByLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
(Ω : Type u_1) →  
(X : Type u_2) →  
  [inst : MeasurableSpace Ω] →  
    [inst_1 : MeasurableSpace X] → MeasureTheory.Measure Ω → MeasureTheory.Measure X → Type (max u_1 u_2)
```

Exact Lean library declaration

```
structure PoissonCountingMeasureByLaw  
  (Ω X : Type*) [MeasurableSpace Ω] [MeasurableSpace X]  
  (P : Measure Ω) (intensity : Measure X) where
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonCountingMeasureByLaw.finiteCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →
{X : Type u_2} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : MeasurableSpace X] →
      {P : MeasureTheory.Measure Ω} →
        {intensity : MeasureTheory.Measure X} →
          (self : EconCSLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) →
            (a : Set X) → (a_1 : Ω) → self.2 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$(\Omega : \text{Type } u_1) \rightarrow [\text{inst} : \text{MeasurableSpace } \Omega] \rightarrow \text{MeasureTheory.Measure } \Omega \rightarrow \text{Type } u_1$

Exact Lean library declaration

```
structure ForwardHomogeneousPoissonCountingProcessByLaw
  (Ω : Type*) [MeasurableSpace Ω] (P : Measure Ω) where
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.count

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →  
  [inst : MeasurableSpace Ω] →  
    {P : MeasureTheory.Measure Ω} →  
      (self : EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) →  
        (a : NNReal) → (a_1 : Ω) → self.4 a a_1
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.intervalCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →  
  [inst : MeasurableSpace Ω] →  
    {P : MeasureTheory.Measure Ω} →  
      (H : EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) →  
        (s t : NNReal) → (ω : Ω) → H.count t ω - H.count s ω
```

Exact Lean library declaration

```
def intervalCount  
  (H : ForwardHomogeneousPoissonCountingProcessByLaw Ω P)  
  (s t : ℝ≥0) (ω : Ω) : ℕ :=  
  H.count t ω - H.count s ω
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.rateExposureParam

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\text{rate } \text{exposure} : \mathbb{R}) \rightarrow (\text{h_nonneg} : 0 \leq \text{rate} * \text{exposure}) \rightarrow \langle \text{rate} * \text{exposure}, \text{h_nonneg} \rangle$

Exact Lean library declaration

```
def rateExposureParam (rate exposure : ℝ)
  (h_nonneg : 0 ≤ rate * exposure) : ℝ≥0 :=
  ⟨rate * exposure, h_nonneg⟩
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
MeasureTheory.IsProbabilityMeasure P →
(rate : ℝ) →
(rate_pos : 0 < rate) →
(count : NNReal → Ω → ℕ) →
(∀ (t : NNReal), Measurable (count t)) →
(∀m (ω : Ω) ∂P, count 0 ω = 0) →
(∀m (ω : Ω) ∂P, Monotone fun t => count t ω) →
ProbabilityTheory.HasIndepIncrements count P →
(∀ {s t : NNReal} (hst : s ≤ t),
ProbabilityTheory.HasLaw (fun ω => count t ω - count s ω)
(ProbabilityTheory.poissonMeasure
(EconCSLib.Probability.PoissonProcess.rateExposureParam rate (↑t - ↑s) ...))
P) →
EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.rate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →  
  [inst : MeasurableSpace Ω] →  
    {P : MeasureTheory.Measure Ω} →  
      (self : EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) → self.2
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.unitIntervalCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →  
  [inst : MeasurableSpace Ω] →  
    {P : MeasureTheory.Measure Ω} →  
      (H : EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) →  
        (n : ℕ) → (omega : Ω) → H.intervalCount (↑n) (↑n + 1) omega
```

Exact Lean library declaration

```
def unitIntervalCount  
  (H : ForwardHomogeneousPoissonCountingProcessByLaw Ω P)  
  (n : ℕ) (omega : Ω) : ℕ :=  
  H.intervalCount (n : ℝ≥0) ((n : ℝ≥0) + 1) omega
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ObservationWindow

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type

Exact Lean library declaration

```
structure ObservationWindow where
  startTime : ℝ
  endTime : ℝ
  start_le_end : startTime ≤ endTime
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ObservationWindow.endTime

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(self : EconCSLib.Probability.PoissonProcess.ObservationWindow) → self.2

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ObservationWindow.startTime

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(self : EconCSLib.Probability.PoissonProcess.ObservationWindow) → self.1

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.ObservationWindow.exposure

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(W : EconCSLib.Probability.PoissonProcess.ObservationWindow) → W.endTime - W.startTime

Exact Lean library declaration

```
def exposure (W : ObservationWindow) : ℝ :=  
  W.endTime - W.startTime
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type

Exact Lean library declaration

```
structure OrderedFiniteJumpTimeline where
  window : ObservationWindow
  count : ℕ
  jumpTime : ℕ → ℝ
  endpoint_mono : Monotone (jumpTimelineEndpoint window.startTime jumpTime)
  last_le_end :
    jumpTimelineEndpoint window.startTime jumpTime count ≤ window.endTime
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.count

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(self : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → self.2

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.jumpTime

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(self : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → (a : ℕ) → self.3 a

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.window

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

(self : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → self.1

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.jumpTimelineEndpoint

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(start : ℝ) →
  (jumpTime : ℕ → ℝ) →
  (a : ℕ) →
    match a with
    | 0 => start
    | i.succ => jumpTime i
```

Exact Lean library declaration

```
def jumpTimelineEndpoint (start : ℝ) (jumpTime : ℕ → ℝ) : ℕ → ℝ
| 0 => start
| i + 1 => jumpTime i
```

-- The `i`th observed interarrival gap in a jump timeline. -/

```
def interarrivalGapFromJumpTimes
  (start : ℝ) (jumpTime : ℕ → ℝ) (i : ℕ) : ℝ :=
  jumpTimelineEndpoint start jumpTime (i + 1) -
  jumpTimelineEndpoint start jumpTime i
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(start : ℝ) →  
  (jumpTime : ℕ → ℝ) →  
    (i : ℕ) →  
      EconCSLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime (i + 1) -  
      EconCSLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime i
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(T : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →  
(j : Fin T.count) → EconCSLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes T.window.startTime T.jumpTime ↑j
```

Exact Lean library declaration

```
def gap (T : OrderedFiniteJumpTimeline) (j : Fin T.count) : ℝ :=  
  interarrivalGapFromJumpTimes T.window.startTime T.jumpTime j.val
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.terminalTailFromJumpTimes

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(start endTime : ℝ) →  
  (jumpTime : ℕ → ℝ) →  
    (count : ℕ) → endTime - EconCSLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime count
```

Exact Lean library declaration

```
def terminalTailFromJumpTimes  
  (start endTime : ℝ) (jumpTime : ℕ → ℝ) (count : ℕ) : ℝ :=  
    endTime - jumpTimelineEndpoint start jumpTime count
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(T : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →  
  EconCSLib.Probability.PoissonProcess.terminalTailFromJumpTimes T.window.startTime T.window.endTime T.jumpTime T.count
```

Exact Lean library declaration

```
def tail (T : OrderedFiniteJumpTimeline) : ℝ :=  
  terminalTailFromJumpTimes T.window.startTime T.window.endTime  
    T.jumpTime T.count
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.interarrival

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow (a : \mathbb{N} \rightarrow \mathbb{R}) \rightarrow a\ n$

Exact Lean library declaration

```
def interarrival (n : ℕ) : (ℕ → ℝ) → ℝ :=  
  fun ω => ω n
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.arrivalTime

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(n : \mathbb{N}) \rightarrow (a : \mathbb{N} \rightarrow \mathbb{R}) \rightarrow \sum i \in \text{Finset.range } (n + 1), \text{EconCSLib.Probability.PoissonProcess.interarrival } i \ a$

Exact Lean library declaration

```
def arrivalTime (n : ℕ) : (ℕ → ℝ) → ℝ :=  
  fun ω => ∑ i ∈ Finset.range (n + 1), interarrival i ω
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.canonicalRenewalCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(t : ℝ) → (a : ℕ → ℝ) → if h : ∃ n, t < EconCSLib.Probability.PoissonProcess.arrivalTime n a then Nat.find h else 0
```

Exact Lean library declaration

```
noncomputable def canonicalRenewalCount (t : ℝ) : (ℕ → ℝ) → ℕ := by
  classical
  exact fun ω => if h : ∃ n : ℕ, t < arrivalTime n ω then Nat.find h else 0
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.exponentialInterarrivalMeasure

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(rate : ℝ) → MeasureTheory.Measure.infinitePi fun x => ProbabilityTheory.expMeasure rate
```

Exact Lean library declaration

```
def exponentialInterarrivalMeasure (rate : ℝ) : Measure (ℕ → ℝ) :=  
  Measure.infinitePi (fun _ : ℕ => ProbabilityTheory.expMeasure rate)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.canonicalForwardCount_hasIndepIncrements

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {rate : ℝ},
  0 < rate →
    ProbabilityTheory.HasIndepIncrements (fun t ω => EconCSLib.Probability.PoissonProcess.canonicalRenewalCount (↑t) ω)
      (EconCSLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate)
```

Exact Lean library declaration

```
theorem canonicalForwardCount_hasIndepIncrements
  {rate : ℝ} (hrate : 0 < rate) :
    ProbabilityTheory.HasIndepIncrements
      (fun (t : ℝ≥0) (ω : ℕ → ℝ) => canonicalRenewalCount (t : ℝ) ω)
      (exponentialInterarrivalMeasure rate)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.isProbabilityMeasure_exponentialInterarrivalMeasure

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {rate : ℝ},  
  0 < rate →  
    MeasureTheory.IsProbabilityMeasure (EconCSLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate)
```

Exact Lean library declaration

```
theorem isProbabilityMeasure_exponentialInterarrivalMeasure  
  {rate : ℝ} (hrate : 0 < rate) :  
    IsProbabilityMeasure (exponentialInterarrivalMeasure rate)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{rate : ℝ} →
(hrate : 0 < rate) →
{ isProbability := ⋯, rate := rate, rate_pos := hrate,
  count := fun t ω => EconCSLib.Probability.PoissonProcess.canonicalRenewalCount (↑t) ω,
  count_measurable :=
    EconCSLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw._proof_1,
  count_zero_ae := ⋯, count_mono_ae := ⋯, hasIndepIncrements := ⋯, increment_hasLaw := ⋯ }
```

Exact Lean library declaration

```
def canonicalForwardHomogeneousPoissonCountingProcessByLaw
  {rate : ℝ} (hrate : 0 < rate) :
  ForwardHomogeneousPoissonCountingProcessByLaw (ℕ → ℝ)
  (exponentialInterarrivalMeasure rate) where
isProbability := isProbabilityMeasure_exponentialInterarrivalMeasure hrate
rate := rate
rate_pos := hrate
count := fun t ω => canonicalRenewalCount (t : ℝ) ω
count_measurable := fun t => measurable_canonicalRenewalCount (t : ℝ)
count_zero_ae := by
  filter_upwards [ae_all_interarrival_positive hrate] with ω hpos
  rw [canonicalRenewalCount_eq_zero_iff]
  left
  simp [arrivalTime] using hpos 0
count_mono_ae := by
  filter_upwards [ae_canonicalRenewalCount_monotone hrate] with ω hmono
  intro s t hst
  exact hmono (NNReal.coe_le_coe.mpr hst)
hasIndepIncrements := canonicalForwardCount_hasIndepIncrements hrate
increment_hasLaw := by
  intro s t hst
  have hs : 0 ≤ (s : ℝ) := NNReal.coe_nonneg s
  have hh : 0 ≤ (t : ℝ) - (s : ℝ) :=
    sub_nonneg.mpr (NNReal.coe_le_coe.mpr hst)
  simp [rateExposureParam] using
    (canonicalRenewalCount_increment_hasLaw_poisson
      (s := (s : ℝ)) (h := (t : ℝ) - (s : ℝ)) hrate hs hh)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.countLikelihood

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(rate exposure : ℝ) → (n : ℕ) → Real.exp (-(rate * exposure)) * (rate * exposure) ^ n / n.factorial
```

Exact Lean library declaration

```
def countLikelihood (rate exposure : ℝ) (n : ℕ) : ℝ :=  
  Real.exp (-(rate * exposure)) * (rate * exposure) ^ n /  
  (n.factorial : ℝ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.cumulativeArrivalVector

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(q : \mathbb{N}) \rightarrow (a : \text{Fin } q \rightarrow \mathbb{R}) \rightarrow (a_1 : \text{Fin } q) \rightarrow \sum j, a \uparrow j$

Exact Lean library declaration

```
def cumulativeArrivalVector (q : ℕ) : (Fin q → ℝ) → Fin q → ℝ :=  
  fun gaps i => ∑ j : {j : Fin q // j ≤ i}, gaps j
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.exponentialBlockDensity

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(rate : ℝ) → (q : ℕ) → (a : Fin q → ℝ) → [] i, ProbabilityTheory.exponentialPDF rate (a i)
```

Exact Lean library declaration

```
def exponentialBlockDensity (rate : ℝ) (q : ℕ) :  
  (Fin q → ℝ) → ℝ≥0∞ :=  
  fun gaps => [] i : Fin q, exponentialPDF rate (gaps i)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.forwardPostStopIntervalCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Ω : Type u_1} →  
  [inst : MeasurableSpace Ω] →  
    {P : MeasureTheory.Measure Ω} →  
      (H : EconCSLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) →  
        (τ : Ω → NNReal) → (u : NNReal) → (a : Ω) → H.intervalCount (τ a) (τ a + u) a
```

Exact Lean library declaration

```
def forwardPostStopIntervalCount  
  {Ω : Type*} [MeasurableSpace Ω] {P : Measure Ω}  
  (H : ForwardHomogeneousPoissonCountingProcessByLaw Ω P)  
  (τ : Ω → ℝ≥0) (u : ℝ≥0) : Ω → ℕ :=  
  fun ω => H.intervalCount (τ ω) (τ ω + u) ω
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.PoissonProcess.noArrivalProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(rate exposure : ℝ) → Real.exp (-(rate * exposure))
```

Exact Lean library declaration

```
def noArrivalProb (rate exposure : ℝ) : ℝ :=  
  Real.exp (-(rate * exposure))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 1567-1570

Verbatim source input

reporting, and misreporting. The latter refers to when residents report problems that, upon inspection by the agency, are not found or are less severe than reported; there is some evidence that such misreporting occurs and is heterogeneous across areas [24].
On the other hand, differential under-reporting is when people, faced with similar problems,

Expanded PaperInterface specification

```
∀ (T : EconCSLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) (D : M.EndpointDensityPresentation)
  {rate : ℝ},
  0 < rate →
  0 < T.window.exposure →
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D rate =
    LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
    LBG24SpatialUnderreporting.sourcePoissonPMF rate T.window.exposure T.count
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.appendix_theorem2_eq8_causal_stopping_source_model

Recorded source-to-Spec screening **Verdict:** not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 1453-1476

Verbatim source input

- [16] Constantine Kontokosta, Boyeong Hong, and Kristi Korsberg. Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain. arXiv preprint arXiv:1710.02452, 2017.
- [17] Constantine E Kontokosta and Boyeong Hong. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. Sustainable Cities and Society, 64:102503, 2021.
- [18] Gabrielle Kruks-Wisner. Seeking the local state: gender, caste, and the pursuit of public services in post-tsunami india. World Development, 39(7):1143–1154, 2011.
- [19] Diane Lambert. Zero-inflated poisson regression, with an application to defects in manufacturing. Technometrics, 34(1):1–14, 1992.
- [20] Myeong Lee, Jieshu Wang, Shawn Janzen, Susan Winter, and John Harlow. Crowdsourcing behavior in reporting civic issues: The case of boston's 311 systems. In Academy of Management Proceedings. Academy of Management Briarcliff Manor, NY 10510, 2021.
- [21] Kristian Lum and William Isaac. To predict and serve? Significance, 13(5):14–19, 2016.
- [22] New York City Emergency Management. Nyc emergency management issues travel advisory tuesday morning through tuesday night, 2020. URL https://www.nyc.gov/site/em/about/press-releases/20200803_pr_nycem-issues-travel-advisory.page. Accessed: 2023-09-01.
- [23] Steven Manson, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles. Ipums national

Expanded PaperInterface specification

```
∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] {P : MeasureTheory.Measure Omega}
(M : LBG24SpatialUnderreporting.LemmalCalendarTimeDurationSourceModel Omega P),
(∀ (t : NNReal), Measurable (M.calendarFirstReports.calendarCount t)) ∧
(∀m (omega : Omega) ∂P, M.calendarFirstReports.calendarCount 0 omega = 0) ∧
(∀m (omega : Omega) ∂P, Monotone fun t => M.calendarFirstReports.calendarCount t omega) ∧
ProbabilityTheory.HasIndepIncrements M.calendarFirstReports.calendarCount P ∧
  ∀ {s t : NNReal},
    s ≤ t →
      ∀ (observedCount : ℕ),
        P.real
          {omega |
            M.calendarFirstReports.calendarCount t omega - M.calendarFirstReports.calendarCount s omega =
              observedCount} =
          EconCSLib.Probability.PoissonProcess.countLikelihood
            (LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity
              M.calendarFirstReports.incidentRate (fun u => ∫ (v : ℝ) in 0..u, M.reportingIntensity v)
              M.durationDensity)
            (↑t - ↑s) observedCount
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.lemmal_calendar_time_duration_source_observed_process

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 1433-1452

Verbatim source input

[11] Yanjun Han and Kirankumar Shiragur. On the competitive analysis and high accuracy optimality of profile maximum likelihood. In Proceedings of the 2021 ACM-SIAM Symposium on Discrete Algorithms (SODA), pages 1317–1336. SIAM, 2021.

[12] Mark O Hill. Local frequency as a key to interpreting species occurrence data when recording effort is not known. Methods in Ecology and Evolution, 3(1):195–205, 2012.

[13] Bryan D Jones, Saadia R Greenberg, Clifford Kaufman, and Joseph Drew. Bureaucratic response to citizen-initiated contacts: Environmental enforcement in detroit. American Political Science Review, 71(1):148–165, 1977.

[14] M K'ery and J Andrew Royle. Hierarchical bayes estimation of species richness and occupancy in spatially replicated surveys. Journal of Applied Ecology, 45(2):589–598, 2008.

[15] Konstantin Klemmer, Daniel B Neill, and Stephen A Jarvis. Understanding spatial patterns in rape reporting delays. Royal Society open science, 8(2):201795, 2021.

Expanded PaperInterface specification

```
∀ {Omega1 : Type u_1} {Omega2 : Type u_2} [inst : MeasurableSpace Omega1] [inst_1 : MeasurableSpace Omega2]
  {P1 : MeasureTheory.Measure Omega1} {P2 : MeasureTheory.Measure Omega2}
  (M1 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega1 P1)
  (M2 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega2 P2),
  M1.reportingRate ≠ M2.reportingRate →
  LBG24SpatialUnderreporting.continuousDurationObservedIncidentRate M1.calendarFirstReports.incidentRate
    M1.reportingRate M1.durationDensity =
  LBG24SpatialUnderreporting.continuousDurationObservedIncidentRate M2.calendarFirstReports.incidentRate
    M2.reportingRate M2.durationDensity →
  (∀m (omega : Omega1) ∂P1,
    Filter.Tendsto (fun n => ↑(M1.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop
      (nhds
        (LBG24SpatialUnderreporting.continuousDurationObservedIncidentRate M1.calendarFirstReports.incidentRate
          M1.reportingRate M1.durationDensity))) ) ∧
  M1.reportingRate ≠ M2.reportingRate ∧
  ∃ observedProcess1 observedProcess2,
    observedProcess1.count = M1.calendarFirstReports.calendarCount ∧
    observedProcess2.count = M2.calendarFirstReports.calendarCount ∧
    observedProcess1.rate = observedProcess2.rate
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.proposition1_calendar_time_homogeneous_source_nonidentifiability

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 1593-1597

Verbatim source input

temporal trends geared more towards an epidemiological study, and Pak et al. [31] use roadway length as part of their proxy. Kontokosta et al. [16] analyze residential building problem complaints in New York City's 311 system—using building conditions as a proxy—and also find that many socioeconomic factors contribute to differential reporting behavior. O'Brien et al. [29] use administrative records from active broken streetlight inspections. While the approach can disentangle

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (M : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P)
  (u : ℝNonNeg),
  ∀ᵐ (ω : Ω) ∂P,
    ((ProbabilityTheory.condExpKernel P (MeasurableSpace.comap M.selection.firstReportTime inferInstance)) ω).real
      {ω' |
        EconCSLib.Probability.PoissonProcess.forwardPostStopIntervalCount M.process M.selection.startTime u ω' = 0} =
      ({ rate := M.rate, rate_pos := ∞ }.measure (Set.Ioi ↑u)).toReal
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.lemma2_forward_selected_start_exponential_tail

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation